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By Amit Katiyar (MCA;JNU)

1. Matching

Column A	Column B
A. $\frac{d}{dx}\cos^{-1}x$	I. $-\frac{1}{1+x^2}$
B . $\frac{d}{dx}\cot^{-1}x$	II. $\frac{1}{\lvert x\rvert\sqrt{x^2-1}}$
C. $\frac{d}{dx}\text{cosec}^{-1}x$	III. $\frac{-1}{\sqrt{1-x^2}}$
D. $\frac{d}{dx}\sec^{-1}x$	IV. $\frac{-1}{\lvert x\rvert\sqrt{x^2-1}}$

(a) (A)-I, (B)-II, (C)-III, (D)-IV
(b) (A)-II, (B)-III, (C)-IV, (D)-I
(c) (A)-III, (B)-IV, (C)-I, (D)-II
(d) (A)-III, (B)-I, (C)-IV, (D)-II

2. Area of region $\{(x,y); x^3 \le y \le x\}$
(a) $\frac{1}{6}$ square unit (b) $\frac{1}{3}$ square unit
(c) $\frac{1}{2}$ square unit (d) $\frac{2}{3}$ square unit

3. Arrange the following the decreasing order.
(A) $\begin{vmatrix} 1 & 3 & 5 \\ 2 & 6 & 10 \\ 31 & 11 & 38 \end{vmatrix}$ (B) $\begin{vmatrix} 67 & 19 & 21 \\ 39 & 13 & 14 \\ 81 & 24 & 26 \end{vmatrix}$
(C) $\begin{vmatrix} 1 & -3 & 2 \\ 4 & -1 & 2 \\ 3 & 5 & 2 \end{vmatrix}$ (D) $\begin{vmatrix} 1 & 4 & 9 \\ 4 & 9 & 16 \\ 9 & 16 & 25 \end{vmatrix}$
(a) A, B, C, D (b) B, A, D, C
(c) C, D, A, B (d) C, A, D, B

4. Which of the following option is false about circles whose equation $x^2+y^2-10x-10y+41=0$, $x^2+y^2-22x-10y+137=0$
(A) circles have same center
(B) circles have no meating time
(C) circles have only one meating time
(D) circles have only two meating time
(a) A, B, C (b) B, C, D
(c) A, B, D (d) A, C, D

5. In university there are total 100 students. 15 offered mathematics only, 12 offered statistics only, 8 offered physics only, 40 offered physics and mathematics, 20 offered physics and statistics, 10 offered mathematics and statistic, 65 offered physics. Tell the number of students not offered any of three.
(a) 1 (b) 2
(c) 3 (d) 4

6. Matching

	Column A		Column B
A.	Neither A nor B	I.	$(A\cap\bar{B})\cup(\bar{A}\cap B)$
B .	Atleast one of A, B or C	II.	$\bar{A}\cap\bar{B}$
C.	Exactly one of A and B	III	A ∪ B ∪ C
D.	All three A, B, C	IV	A ∩ B ∩ C

(a) (A)-I, (B)-II, (C)-III, (D)-IV
(b) (A)-II, (B)-III, (C)-I, (D)-IV
(c) (A)-II, (B)-I, (C)-IV, (D)-III
(d) (A)-II, (B)-IV, (C)-I, (D)-III

7. Accurance of an event is usually signaled by interrupt from software by actuating special operation called-
(a) system call
(b) Bootstrap program
(c) Service routine
(d) interrupt vector

8. Mean and variance of 12 observation are $\frac{9}{2}$ and 4 respectively later it was observed that two observation work consider 9 and 10 instead of 7 and 14 respectively. If correct variance is $\frac{m}{n}$ where m and n are coprime then m + n is
(a) 316 (b) 315
(c) 314 (d) 317

9. Which of the following is false about
$\lim_{y\to 0}\frac{\sqrt{1+\sqrt{1+y^4}-\sqrt{2}}}{y^4}$
(A) limit exist and equal to $\frac{1}{4\sqrt{2}}$
(B) limit does not exist
(C) limit exist and equal to $\frac{1}{2\sqrt{2}}$
(D) limit exist and equal to $\frac{1}{2\sqrt{2}(\sqrt{2}+1)}$
(a) A, B, C (b) B, C, D
(c) A, B, D (d) A, C, D

10. Arrange the following in sequence find variance of individual observation
(A) compute $\bar{X}$ of observation $x_1, x_2 \dots x_n$
(B) square the deviation obtain sum $\sum_{i=1}^{n}(x_i - \bar{X})^2$
(C) devide the sum $\sum_{i=1}^{n}(x_i - \bar{X})^2$ by n
(D) take deviation of observation from mean that is $(x_i - \bar{X})$ where i = 1, 2, 3 .....
(a) A, B, C, D (b) A, D, B, C
(c) A, C, B, D (d) B, C, D, A

11. Arrange the parabola in increasing order of length of their latus rectum
(A) $y^2 = 8x$
(B) $4x^2 + y = 0$
(C) $y^2 - 4y - 3x + 1 = 0$
(D) $y^2 - 4y + 4x = 0$
(a) A, B, C, D (b) B, C, A, D
(c) C, B, A, D (d) B, C, D, A

12. Which of the following is correct
(A) in directed graph sum of length of all adjancy list is $|E|$
(B) the adjancy matrix requires O (V^2) memory
(C) In an undirected graph the sum of length of all adjancy list is $|E|$
(D) the memory requirement of adjancy list is depending on number of edge in graph
(a) A, B, C (b) B, C, D
(c) A, B, D (d) A, C, D

13. Matching

	Column A		Column B
A.	$(\sqrt{2}+1)+1+(\sqrt{2}\mp 1)+ \dots\dots\dots \infty$	I.	$\frac{19}{24}$
B.	$\frac{1}{2}+\frac{1}{3^2}+\frac{1}{2^3}+\frac{1}{3^4}+\frac{1}{2^5}+\frac{1}{3^6}+ \dots\dots\dots \infty$	II.	6
C.	$6^{\frac{1}{2}} \times 6^{\frac{1}{4}} \times 6^{\frac{1}{8}} \dots\dots \infty$	III	$8(2+\sqrt{2})$
D.	$8+4\sqrt{2} + 4 + \dots\dots \infty$	IV	$\frac{4+3\sqrt{2}}{2}$

(a) (A)-I, (B)-II, (C)-III, (D)-IV
(b) (A)-IV, (B)-I, (C)-II, (D)-III
(c) (A)-IV, (B)-I, (C)-III, (D)-II
(d) (A)-I, (B)-IV, (C)-III, (D)-II

14. S, T are symmetric matrix of same order then which of the following statement are true
(A) S + T is symmetric matrix
(B) ST - TS is skew symmetric matrix
(C) ST + TS is symmetric matrix
(D) ST - TS is symmetric matrix
(a) A, B, C (b) B, C, D
(c) A, B, D (d) A, C, D

15. Matching

	Column A		Column B
A.	BFS	I.	Shortest Part
B.	Sparse graph	II.	Adjancy list representation
C.	Collision Resolution	III	Open Addressing
D.	Link list	IV	Self Referential structure

(a) (A)-I, (B)-II, (C)-III, (D)-IV
(b) (A)-I, (B)-II, (C)-IV, (D)-III
(c) (A)-II, (B)-I, (C)-III, (D)-IV
(d) (A)-IV, (B)-III, (C)-II, (D)-I

16. Consider Procedure list search (L, K) find the first element with key K in list L by simple linear search, returning pointer to its elements. Arrange the correct order.
(A) x = L . Head
(B) while x ≠ Nil and x . key ≠ K
(C) return x
(D) x = x . next
(a) A, B, D, C (b) B, C, D, A
(c) A, D, B, C (d) A, B, C, D

17. The curve x = $y^2, xy = k$ cut at right angle. Find the value of k
(a) $\frac{1}{2\sqrt{2}}$ (b) 2
(c) 4 (d) $\frac{1}{8}$

18. The foci of hyperbola coincide with foci of ellipse $\frac{x^2}{25}+\frac{y^2}{9} = 1$. So equation of hyperbola if e = 2
(a) $\frac{x^2}{4}-\frac{y^2}{12} = 1$ (b) $\frac{x^2}{16}-\frac{y^2}{12} = 1$
(c) $\frac{x^2}{12}-\frac{y^2}{4} = 1$ (d) $\frac{x^2}{2}-\frac{y^2}{9} = 1$

19. Semaphore operation in sequence for preducer processes in bounded buffer problem using semaphore empty, full, mutex
(A) Singal (Full) → indicating that the new item has been preduce and added to the buffer
(B) Wait (Empty) → check if there is space in the buffer to add new item
(C) Product an item and store it in buffer
(D) Wait (Mutex) → center the critical section to insert the item safely
(E) Signal (mutex) → leaves the critical section after inserting the item
(a) A, E, C, B, D (b) C, B, A, D, E
(c) D, E, A, C, B (d) B, D, C, E, A

20. Which of the following have same image
(A) J (B) K (C) A (D) H
(a) C, D (b) A, D
(c) A, C (d) B, D

21. Tree used to represent arithmetic expression is called _____ tree.
(a) Threaded (b) Game
(c) Red - Black (d) expression

22. Matching

	Column A		Column B
A.	AND	I.	output 1 only if both input 1
B.	OR	II.	output 1 if any input 1
C.	NAND	III	output 0 only if both input 1
D.	NOR	IV	output 1 only if both input 0

(a) (A)-IV, (B)-III, (C)-II, (D)-I
(b) (A)-I, (B)-III, (C)-IV, (D)-II
(c) (A)-I, (B)-II, (C)-III, (D)-IV
(d) (A)-II, (B)-I, (C)-IV, (D)-III

23. Fundamental function of register in digital circuit is to
(a) perform arithmetic operation
(b) store binary information
(c) compared two number
(d) encode input signal

24. Arrange the following in increasing time complexity.
(A) Linear search worst case
(B) binary search best case
(C) Quick sort expected running time
(D) bubble sort worst case
(a) B, A, C, D (b) C, B, A, D
(c) D, E, A, C (d) B, D, C, E

25. In _____ linked list the last node is link field point to first node of list.
(a) Circular (b) Linear
(c) Sequential (d) Indexed

26. Matching

	Column A		Column B
A.	Counter register	I.	Read by host to get input
B.	Status register	II.	Written by host to sent output
C.	Data in register	III	Contain bit that can be read by host
D.	Data out register	IV	Can be written by host to start command

(a) (A)-I, (B)-II, (C)-III, (D)-IV
(b) (A)-IV, (B)-III, (C)-I, (D)-II
(c) (A)-III, (B)-IV, (C)-II, (D)-I
(d) (A)-I, (B)-II, (C)-IV, (D)-III

27. Let f be function $f(x) = \begin{cases} \frac{\tan x}{x} & x \neq 0 \\ 1 & x = 0 \end{cases}$
Assertion: x = 0 is point of minima of f
Reason: $f'(0) = 0$
(a) Assertion is true, Reason is true, and Reason is the correct explanation of the Assertion.
(b) Assertion is true, Reason is true, but Reason is not the correct explanation of the Assertion.
(c) Assertion is true, but Reason is false.
(d) Assertion is false, but Reason is true.

28. In _____ search we begin search at staring node and when we come across dead end then we back track.
(a) connected (b) bubble
(c) breadth first (d) depth first

29. $\int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{1+\sin x \cos x} dx$

(a) $\frac{\pi}{2\sqrt{3}}$ (b) $\frac{\pi}{2\sqrt{2}}$

(c) $\frac{\pi}{9}$ (d) $\frac{\pi}{\sqrt{3}}$

30. **Assertion:** Deadlock accur in multi programming when each thread in set is waiting for event that can be caused only by another thread in same set.

Reason: A thread must always request before using it and realise it after using it. If requested resourse are not available thread enter a waiting state until resource become free.

(a) Assertion is true, Reason is true, and Reason is the correct explanation of the Assertion.

(b) Assertion is true, Reason is true, but Reason is not the correct explanation of the Assertion.

(c) Assertion is true, but Reason is false.

(d) Assertion is false, but Reason is true.

31. Study the given information carefully and answer the question:

There are seven books, one each of Psychology, Hindi, English, Sociology, Economics, Education and Accountancy lying on a table one above the other. Sociology is on the top of all books countancy is immediately below cation which is immediate below Sociology. Economics is immediately above Psychology but not in the middle. Hindi is immediately below Psychology.

Which three books are between Accountancy and Hindi?

(a) English, Economics and Psychology

(b) Economics, Psychology and Education

(c) Economics, Psychology and Hindi

(d) Cannot be determined

32. select the odd one out

(a) crow (b) pigeon

(c) Butterfly (d) Parrot

33. Assertion: the leaves of plant are green

Reason: its because of Chromoplast the green pigment

(a) Assertion is true, Reason is true, and Reason is the correct explanation of the Assertion.

(b) Assertion is true, Reason is true, but Reason is not the correct explanation of the Assertion.

(c) Assertion is true, but Reason is false.

(d) Assertion is false, but Reason is true.

34. A target thread periodically check whether it should terminate allowing it an opportunity to terminate it self in an orderly fashion ?

(a) synchronous cancellation

(b) A synchronous cancellation

(c) default cancelation

(d) deferred cancelation

35. Matching

Column A		Column B
A. 4 : 17 : : 7 : ?	I.	18
B. 720 : 840 : : 60 : ?	II.	82
C. 225 : 256 : : 289 : ?	III.	324
D. 21 : 3 : : 574 : ?	IV.	50

(a) (A)-I, (B)-II, (C)-III, (D)-IV

(b) (A)-IV, (B)-I, (C)-III, (D)-II

(c) (A)-III, (B)-IV, (C)-II, (D)-I

(d) (A)-I, (B)-II, (C)-IV, (D)-III

36. **Statement :** I. All windows are door.
II. No door is a wall

Conclusion : I. No window is wall.
II. No wall is door

(a) neither I nor II

(b) only I

(c) only II

(d) Both I and II

37. Which of the following condition should be full filled in critical section.

(A) Mutual exclusion (B) Race condition

(C) Progress (D) Bounded wait

(a) A, C, D (b) B, C, D

(c) A, B, D (d) A, B, C, D

38. Matching

	Column A		Column B
A.	Decoder	I.	Selecet input and output of many
B .	Multiplexer	II.	convert binary input into 2^n output
C.	Encoder	III	convert 2^n input into binary output
D.	Demultiplexer	IV	route one input to several output

(a) (A)-I, (B)-II, (C)-III, (D)-IV
(b) (A)-II, (B)-I, (C)-III, (D)-IV
(c) (A)-III, (B)-II, (C)-IV, (D)-I
(d) (A)-IV, (B)-III, (C)-II, (D)-I

39. Assertion: $\frac{1}{3}, -\frac{1}{2}, \frac{3}{4}, -\frac{9}{8}$ form an AP
Reason: the constant sequence is only sequence which is both AP as well as GP
(a) Assertion is true, Reason is true, and Reason is the correct explanation of the Assertion.
(b) Assertion is true, Reason is true, but Reason is not the correct explanation of the Assertion.
(c) Assertion is true, but Reason is false.
(d) Assertion is false, but Reason is true.

40. Which of the following is a universal logic gate.
(a) AND (b) OR
(c) NAND (d) XOR

41. The condition for which JK flip flop gives T flip flop
(a) J = 0 , K =1 (b) J = 1, K = 1
(c) J = 1, K = 0 (d) J = 0, K = 0

42. find the ? in the given series
240, ____, 120, 40, 10, 2
(a) 180 (b) 240
(c) 360 (d) None

43. in memory management which of the following allows non contiguous memory allocation
(a) paging (b) address binding
(c) variable (d) Partition

44. Matching

	Column A		Column B
A.	Taxonomy	I.	Biography
B.	Pedology	II.	History
C.	Man	III.	Social
D.	Nation	IV.	classification

(a) A-IV, B-I, C-III, D-II
(b) A-IV, B-II, C-I, D-III
(c) A-I, B-IV, C-II, D-III
(d) A-III, B-I, C-IV, D-II

45. Half adder has
(a) two input two output
(b) one input two output
(c) two input one output
(d) one input one output

46. Maching

(A)	FCFS	I.	Give CPU according to highest priority
(B)	SJF	II.	Gives a CPU to proses which come first
(C)	Priority Scheduling	III.	Gives CPU to process with shortest burst time
(D)	RR	IV.	Time shairing

(a) (A)-II, (B)-III, (C)-I, (D)-IV
(b) (A)-I, (B)-II, (C)-III, (D)-IV
(c) (A)-II, (B)-I, (C)-III, (D)-IV
(d) (A)-IV, (B)-III, (C)-II, (D)-I

47. Find the missing ?
M, N, O, L, R, J, V, ?
(a) A (b) E
(c) F (d) H

48. **Assertion:** Binary search tree property allow to print all keys in binary search tree in sorted order by simple recursive algorithm called inorder tree walk.
Reason: The inorder tree walk algorithm print the key root of subtree between printing value in its left subtree and printing those in its right subtree.
(a) Assertion is true, Reason is true, and Reason is the correct explanation of the Assertion.
(b) Assertion is true, Reason is true, but Reason is not the correct explanation of the Assertion.
(c) Assertion is true, but Reason is false.
(d) Assertion is false, but Reason is true.

49. **Assertion:** A (BA) and $(AB)A$ are symmetric matrices.

Reason: AB is symmetric matrix, if matrix multiplication of A with B is commutation

(a) Assertion is true, Reason is true, and Reason is the correct explanation of the Assertion.
(b) Assertion is true, Reason is true, but Reason is not the correct explanation of the Assertion.
(c) Assertion is true, but Reason is false.
(d) Assertion is false, but Reason is true.

50. Arrange the following steps of retrieving data from memory unit.

(A) Memory Address Register send address to memory chip
(B) The address is placed on address bus
(C) The control unit sends signal
(D) Data is retrieved and place into the Memory Data Register

(a) D, C, A, B (b) A, B, C, D
(c) A, C, D, B (d) B, D, A, C

51. Maching

(A)	Minimum Spanning Tree	I.	Height balanced
(B)	Topological Sort	II.	Stack
(C)	Recursion	III.	Kruskal's algorithm
(D)	AVL Tree	IV.	DFS

(a) (A)-I, (B)-II, (C)-III, (D)-IV
(b) (A)-II, (B)-I, (C)-III, (D)-IV
(c) (A)-IV, (B)-III, (C)-II, (D)-I
(d) (A)-III, (B)-IV, (C)-II, (D)-I

52. A matrix is both symmetric and Skew symmetric matrix

(a) A is diagonal matrix
(b) A is zero matrix
(c) A is scalar matrix
(d) A is square matrix

53 Cub related to Tiger then Fawn related to _____

(a) Monkey (b) Stag
(c) Ass (d) Sheep

54. Which algorithm have same order of complexity

A. Deletion from Queue
B. Worst case search in binary search tree
C. Insertion in stack
D. Quick sort worst case

(a) C, D (b) A, C
(c) A, B (d) B, C

55. Complement of $\overline{A+B}$

(a) $\bar{A}+\bar{B}$ (b) $\bar{A}.\bar{B}$
(c) A + B (c) A.B

56. Which of the following are combinational circuit

A. Half adder B. SR flip flop
C. Full adder D. D flip flop

(a) A, C (b) B, D
(c) B, C, D (d) A, B, C, D

57. Length or number of data element of Array can be obtained from index set by which of the following formulae.

(a) Lower bound - Upper bound - 1
(b) Lower bound - Upper bound + 1
(c) Upper bound - Lower bound + 1
(d) Upper bound + Lower bound - 1

58. Cooperating process required an interprocess communication mechanism. This work on which of the following fundamental model.

(A) Shared Memory
(B) Cascading termination
(C) Message passing
(D) Cooperating

(a) C, D (b) A, C
(c) A, D (d) B, D

59. Equation of pair of lines y = px, y = qx written as $(y-px)(y-qx)=0$ equation of pair of angle bisector of lines $x^2-4xy-5y^2=0$

(a) $x^2-3xy+y^2=0$
(b) $x^2+3xy-y^2=0$
(c) $x^2+4xy-y^2=0$
(d) $x^2-3xy-y^2=0$

60. Which of the following is not an application of stack.
(a) Tower of Hanoi
(b) evaluation of postfix expression
(c) Priority queue
(d) Recurssion

61. Arrange the following steps of K-map simplification in correct order.
(A) Write minimized SOP / POS expression
(B) Plot truth table outputs (0's and 1's) on K-map
(C) Identify prime implicants and essential prime implicants
(D) Draw groups (loops) of adjacent 1's or 0's that are powers of 2.
(a) A, B, C, D (b) C, A, B, D
(c) B, C, A, D (d) B, D, C, A

ANSWER KEY

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
d	c	d	c	a	b	c	d	b	b
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
d	c	b	a	a	a	a	a	d	a
21.	22.	23.	24.	25.	26.	27.	28.	29.	30.
d	c	b	a	a	b	b	d	a	b
31.	32.	33.	34.	35.	36.	37.	38.	39.	40.
a	c	c	d	b	d	a	b	d	c
41.	42.	43.	44.	45.	46.	47.	48.	49.	50.
b	b	a	a	a	a	d	a	b	b
51.	52.	53.	54.	55.	56.	57.	58.	59.	60.
d	b	b	b	c	a	c	b	b	c
61.									
d									